# `genQE` — Two-Nucleon GCF Event Generator

Physics & Pipeline Documentation

GCF with 3N Project

October 22, 2026

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# 1 Overview

`genQE` is a Monte Carlo event generator for quasi-elastic (QE) electron scattering off nuclei within the **Generalized Contact Formalism (GCF)**. It generates events of the type

$$e + A \longrightarrow e' + N_{\text{lead}} + N_{\text{rec}} + (A-2)^*, \tag{1}$$

where the electron scatters off one nucleon of a short-range correlated (SRC) pair. The struck nucleon ($N_{\text{lead}}$) carries the virtual-photon momentum; the partner ($N_{\text{rec}}$) recoils, and the residual ($A-2$) system is left with excitation energy $E^*$.

The generator factorizes the problem into two independent parts:

1. **Nuclear structure:** sampling the SRC pair's internal momenta via the GCF (contact coefficients $\times$ universal functions).
2. **Scattering dynamics:** computing the electron–nucleon cross section at the sampled kinematics.

# 2 Generalized Contact Formalism

## 2.1 Contact factorization

The GCF expresses the two-body momentum distribution of a nucleus $A$ as

$$n_A^{(ij)}(\boldsymbol{k}_{\text{rel}}) \;=\; C_{ij}^A \left|\varphi_{ij}(k_{\text{rel}})\right|^2, \tag{2}$$

where $i, j$ label isospin–spin channels, $C_{ij}^A$ are the *nuclear contact coefficients* (nucleus-dependent, extracted from *ab initio* calculations or data), and $\varphi_{ij}(k)$ are *universal* (nucleus-independent) functions determined by the short-range behavior of the chosen NN potential.

Four channels are implemented:

| Channel | Pair | Quantum numbers |
|---|---|---|
| `pp0` | proton–proton | $^1S_0$ |
| `nn0` | neutron–neutron | $^1S_0$ |
| `pn0` | proton–neutron | $^1S_0$ |
| `pn1` | proton–neutron | $^3S_1$ (deuteron-like, dominant) |

For identical-nucleon pairs ($pp$ or $nn$) the spectral function used is

$$S(k) = 2\,C\;\varphi^2(k), \tag{3}$$

while for $pn$ pairs it is

$$S(k) = C_{pn0}\,\varphi_{pn0}^2(k) + C_{pn1}\,\varphi_{pn1}^2(k). \tag{4}$$

The universal functions $\varphi^2(k)$ are tabulated in 100 bins covering $k \in [0,\, 10]\,\text{fm}^{-1}$ with $\Delta k = 0.1\,\text{fm}^{-1}$ and linearly interpolated at runtime.

## 2.2 Available NN interactions

Several sets of $\varphi^2(k)$ and contact values are included:

- **AV18** — Argonne $v_{18}$ (default)
- **N2LO_10** — Chiral N$^2$LO, $R = 1.0\,\text{fm}$
- **N2LO_12** — Chiral N$^2$LO, $R = 1.2\,\text{fm}$
- **N3LO_600** — Chiral N$^3$LO, $\Lambda = 600\,\text{MeV}$ (tables only: no contacts, so it aborts)
- **AV4Pc** — Argonne AV4′
- **NV2_1a** — Norfolk NV2+Ia
- **AV18_deut** — AV18 with deuteron-specific wave function (no contact dependence)

### 2.3 Contact coefficients

Contact values $C_{ij}^{A}$ are stored per nucleus and NN interaction. They are loaded by `gcfSRC` on initialization. Representative values (AV18, $^{12}$C):

$$C_{pp0} \approx C_{nn0} \approx 5.4, \qquad C_{pn0} \approx 5.4, \qquad C_{pn1} \approx 110.7. \tag{5}$$

The $pn1$ (deuteron-like) channel dominates by roughly an order of magnitude.

## 3 Nuclear Parameters

Each target nucleus is described by a `gcfNucleus` object that stores:

- $A$, $Z$, nuclear mass $m_A$
- $\sigma_{\mathrm{CM}}$: width (GeV/$c$) of the Gaussian center-of-mass momentum distribution of the pair
- $E^{*}_{\mathrm{max}}$: maximum excitation energy of the $(A-2)$ residual
- $(A-2)$ masses: $m^{pp}_{A-2}$, $m^{pn}_{A-2}$, $m^{nn}_{A-2}$ (depend on which pair is removed)

| Nucleus | $A$ | $\sigma_{\mathrm{CM}}$ (GeV/$c$) | $E^{*}_{\mathrm{max}}$ (GeV) | Note |
|---|---|---|---|---|
| $^{2}$H | 2 | 0 | 0 | No CM motion, no residual |
| $^{3}$He | 3 | 0.10 | 0 | $(A-2)$ = single nucleon |
| $^{4}$He | 4 | 0.10 | 0.010 | $(A-2)$ = deuteron (pn) or $2N$ (pp/nn) |
| $^{12}$C | 12 | 0.15 | 0.030 | |
| $^{56}$Fe | 56 | 0.15 | 0.050 | |
| $^{208}$Pb | 208 | 0.15 | 0.100 | |

The excitation energy can be sampled from a Gaussian $E^{*} \sim \mathcal{N}(E^{*}_{\mathrm{mean}}, \sigma_E)$ or held fixed.

## 4 Event Generation Pipeline

### 4.1 Step 1: Pair type selection

A pair type $(ij) \in \{pp, nn, pn\}$ is chosen uniformly. The event weight carries an overall factor of 4 (2 nucleons $\times$ 2 possible isospin assignments).

### 4.2 Step 2: Decay function — sampling the SRC pair

The SRC pair's kinematics are generated in `gcfGenerator::decay_function()`.

**Center-of-mass momentum.** Each Cartesian component of $\boldsymbol{P}_{\mathrm{CM}}$ is sampled independently from a Gaussian:

$$P_{\mathrm{CM},\alpha} \sim \mathcal{N}(0,\ \sigma_{\mathrm{CM}}). \tag{6}$$

**Relative momentum.** The relative momentum $\boldsymbol{k}_{\mathrm{rel}}$ is sampled uniformly:

$$k \in [k_{\mathrm{min}},\, k_{\mathrm{max}}], \quad \cos\theta_{\mathrm{rel}} \in [-1,\, 1], \quad \phi_{\mathrm{rel}} \in [0,\, 2\pi]. \tag{7}$$

Default range: $k_{\mathrm{min}} = 0.2\,\mathrm{GeV}/c$, $k_{\mathrm{max}} = 1.05\,\mathrm{GeV}/c$.

**Weight from spectral function.** The phase-space weight is

$$w_{\mathrm{decay}} = \Delta k\ \Delta(\cos\theta)\ \Delta\phi\ \times\ \frac{k^2\, S(k,\, i, j)}{(2\pi)^3}, \tag{8}$$

where $S(k)$ is the GCF spectral function from Eq. (2).

**Pauli blocking cut.** If $k < k_{\text{cut}}$ (default $0.25\,\text{GeV}/c$; 0 for deuterium), the event is rejected ($w = 0$).

**Individual momenta.**

$$\boldsymbol{p}_{\text{lead}} = \tfrac{1}{2}\boldsymbol{P}_{\text{CM}} + \boldsymbol{k}_{\text{rel}}, \qquad \boldsymbol{p}_{\text{rec}} = \tfrac{1}{2}\boldsymbol{P}_{\text{CM}} - \boldsymbol{k}_{\text{rel}}. \tag{9}$$

**Lead nucleon energy.** From energy conservation:

$$E_{\text{lead}} = m_A - E_{A-2} - E_{\text{rec}}, \qquad E_{\text{rec}} = \sqrt{|\boldsymbol{p}_{\text{rec}}|^2 + m_N^2}, \tag{10}$$

where $E_{A-2} = \sqrt{|\boldsymbol{p}_{A-2}|^2 + m_{A-2}^2}$ and $\boldsymbol{p}_{A-2} = -\boldsymbol{P}_{\text{CM}}$.

### 4.3 Step 3: Light-cone variant

An alternative light-cone kinematic mode (`decay_function_lc`) is available. It uses light-cone momentum fractions

$$\alpha = \frac{p^+}{\bar{m}}, \qquad \bar{m} = m_A/A, \tag{11}$$

with transverse momentum $\boldsymbol{p}_\perp$. The relative light-cone momentum fraction $\alpha_{\text{rel}}$ is sampled uniformly, and the constraint

$$k_{\min} = \frac{|1 - \alpha_{\text{rel}}|}{\sqrt{\alpha_{\text{rel}}(2 - \alpha_{\text{rel}})}}\, m_N \tag{12}$$

is enforced to keep kinematics physical. The weight is

$$w_{\text{lc}} = \Delta\alpha\ \Delta k\ \Delta\phi\ \times\ \frac{k\sqrt{k^2 + m_N^2}\ S(k)}{(2\pi)^3}. \tag{13}$$

### 4.4 Step 4: Electron scattering vertex

Given the lead nucleon's 4-momentum from Step 2, the scattered electron kinematics are determined.

**Sampling.** Two variables are drawn uniformly:

$$Q^2 \in [Q^2_{\min},\, Q^2_{\max}], \qquad \phi_k \in [0,\, 2\pi]. \tag{14}$$

**Kinematic constraint.** The electron's transverse momentum $k_\perp$ is determined by solving a quadratic arising from energy–momentum conservation of the full reaction. Let $p_1^- = E_1 - p_{1z}$ denote the nucleon's light-cone minus component and $k^- = Q^2/(2E_{\text{beam}})$. Then

$$A_e = \frac{k^-}{p_1^-}, \qquad c_e = A_e\big(p_1^+\, k^- - 2E_{\text{beam}}\, p^-_{\text{lead}} - \text{virt}\big), \qquad b_e = -2A_e\, y, \tag{15}$$

where $y = p_{1\perp}\cos(\phi_k - \phi_1)$ and "virt" encodes off-shell effects. The solution is

$$k_\perp = \frac{-b_e \pm \sqrt{b_e^2 - 4c_e}}{2}. \tag{16}$$

Events with $D = b_e^2 - 4c_e < 0$ are rejected. When two physical solutions exist, both are equally probable and the weight is multiplied by 2.

**Jacobian.** The $\delta$-function constraint contributes a Jacobian factor

$$J = 2\,E_{\rm beam}\,E_k\left|1 - \frac{p_{1z} + y\tan(\theta_k/2) + E_{\rm beam} - E_k}{E_{\rm lead}}\right|, \qquad w \;*=\; 1/J. \tag{17}$$

### 4.5 Step 5: Cross section

The electron–nucleon cross section $\sigma_{eN}$ is evaluated at the sampled kinematics. Three prescriptions are available:

**On-shell.** Standard Rosenbluth formula assuming the nucleon is on-shell:

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_{\rm beam}^2\sin^4(\theta/2)}\,\frac{E_k}{E_{\rm beam}}\left[\frac{G_E^2 + \tau G_M^2}{1+\tau}\cos^2\frac{\theta}{2} + 2\tau G_M^2\sin^2\frac{\theta}{2}\right], \tag{18}$$

with $\tau = Q^2/(4m_N^2)$.

**CC1 (de Forest, default).** Off-shell generalization using current-conservation prescription 1 (T. de Forest, Nucl. Phys. A **392**, 232 (1983), doi:10.1016/0375-9474(83)90124-0). Constructs structure functions $w_C$, $w_T$, $w_S$, $w_I$ from form factors $G_E$, $G_M$, and evaluates

$$\sigma_{\rm CC1} = \sigma_{\rm Mott}\left[v_L\,w_C + v_T\,w_T + v_S\,w_S + v_I\,w_I\right], \tag{19}$$

where $v_{L,T,S,I}$ are kinematic response functions.

**CC2.** Alternative current-conservation prescription with modified structure function weighting (different treatment of $\bar{p}$ components).

### 4.6 Step 6: Form factors

Three electromagnetic form-factor models:

**Dipole.** $G_E^p = G_D = (1 + Q^2/0.71)^{-2}$, $G_M^p = \mu_p\,G_D$.

**Kelly (default).** Rational parameterization (J. J. Kelly, Phys. Rev. C **70**, 068202 (2004), doi:10.1103/PhysRevC.70.068202):

$$G(Q^2) = \frac{1 + a_1\tau}{1 + b_1\tau + b_2\tau^2 + b_3\tau^3}. \tag{20}$$

Fit parameters given separately for $G_E^p$, $G_M^p/\mu_p$, $G_E^n$, $G_M^n/\mu_n$.

**Ye.** 13-parameter $z$-expansion fit of Z. Ye *et al.* (doi:10.1016/j.physletb.2017.11.023; Phys. Lett. B **777**, 8 (2018)).

### 4.7 Step 7: Final weight

The total event weight is

$$w = 4\;\times\; w_{\rm decay}\;\times\;\left(\Delta Q^2\cdot 2\pi\right)\;\times\;\frac{1}{J}\;\times\;\sigma_{eN}. \tag{21}$$

### 4.8 Step 8: Optional corrections

**Coulomb correction.** The electron energy is shifted before and after scattering by the nuclear Coulomb potential:

$$\Delta E_{\text{Coulomb}} = \frac{3}{2}\,\frac{Z\alpha_{\text{em}}}{R_C}, \qquad R_C = 1.1\,A^{1/3}\,\text{fm}. \tag{22}$$

**Bremsstrahlung.** Radiative corrections via the peaking approximation. A Poisson-distributed number of photons is emitted; each photon's energy fraction sampled from

$$P(\Delta E/E) \propto \left(\frac{\Delta E}{E}\right)^{\lambda-1}, \qquad \lambda = \frac{\alpha}{\pi}\left(\ln\frac{4E^2}{m_e^2} - 1\right). \tag{23}$$

## 5 Output

Events are stored in a ROOT `TTree` named `genT` with branches:

| Branch | Type | Description |
|---|---|---|
| `lead_type` | int | PDG code (2212 = $p$, 2112 = $n$) |
| `rec_type` | int | PDG code of recoil nucleon |
| `pe[3]` | double | Scattered electron $\boldsymbol{p}_{e'}$ |
| `pLead[3]` | double | Lead nucleon $\boldsymbol{p}_{\text{lead}}$ |
| `pRec[3]` | double | Recoil nucleon $\boldsymbol{p}_{\text{rec}}$ |
| `pAm2[3]` | double | $(A-2)$ momentum |
| `pRel[3]` | double | Relative momentum $\boldsymbol{k}_{\text{rel}}$ |
| `q[3]` | double | Momentum transfer $\boldsymbol{q}$ |
| `weight` | double | Event weight |

## 6 Summary of Algorithm

For each Monte Carlo trial:

1. Select pair type ($pp$, $nn$, or $pn$) uniformly.
2. Sample $\boldsymbol{P}_{\text{CM}}$ (Gaussian) and $\boldsymbol{k}_{\text{rel}}$ (uniform in $k$, $\cos\theta$, $\phi$).
3. Evaluate GCF spectral function $S(k)$; apply Pauli cut.
4. Derive individual nucleon momenta and $(A-2)$ recoil.
5. (Optional) Apply Coulomb shift and bremsstrahlung to beam.
6. Sample $Q^2$ and $\phi_k$; solve quadratic for $k_\perp$.
7. Compute Jacobian $J$ from energy conservation constraint.
8. Evaluate $\sigma_{eN}$ (CC1/CC2/on-shell with Kelly/dipole/Ye).
9. Multiply all weight factors; store event if $w > 0$.